Student Name (in English): _____ Student ID: _____

Question 1: **Multiple choices** To be answered in (30) Minutes [ ] / 10 Marks

Choose one correct answer for each statement and fill the following table:

Question	1	2	3	4	5	6	7	8	9	10
Answer										

1. **Merge sort algorithm follows _____ design technique:**
   a. Incremental
   b. Greedy
   c. Divide and conquer
   d. All the above answers are incorrect

2. **Suppose you want to sort an array of *n* integers in the range [0..n^3] (k= n^3) using *counting sort*. Then, the running time of the counting sort will be:**
   a. n
   b. n logn
   c. n^3
   d. n^4

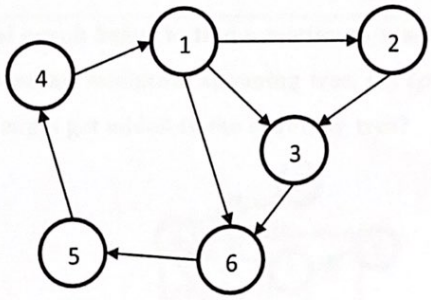
3. **Select the correct statement about BFS and DFS algorithms:**
   a. Time complexity of DFS is lower while the space complexity is the same.
   b. Time complexity of BFS is lower while the space complexity is the same
   c. Space complexity of DFS is lower while the time complexity is the same.
   d. Space complexity of BFD is lower while the time complexity is the same

4. **Consider the list of integers: 365, 6443, 642, 223, 532, 2, 432, the output of the list after the first pass of the radix sort algorithm?**
   a. 642, 532, 2, 432, 6443, 223, 365
   b. 365, 6443, 642, 223, 532, 2, 432
   c. 223, 2, 365, 432, 532, 6443, 642
   d. All the above the answers are incorrect

5. **Assumes we need to sort a list of 5 integer numbers that uniformly distributed over [0, 1) using bucket sort, then we need to create _____ buckets.**
   a. 1
   a. 5
   b. 10
   c. Cannot be determined

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Based on the graph below, answer the following questions:



6. **The path: 1, 2, 3, 6, 5, 4 is:**
   a. Cycle
   b. Circuit
   c. Hamiltonian Cycle
   d. All the above answers are incorrect

7. **The in-degree of vertex 6 is:**
   a. 1
   b. 2
   c. 3
   d. All the above answers are incorrect

8. **Choose the correct statement:**
   a. The path: 1, 2, 3, 1 is simple
   b. Vertex 6 is reachable from vertex 5
   c. The graph is Unconnected graph
   d. All the above answers are correct

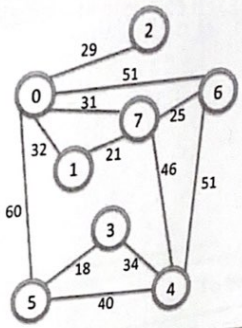
9. **The depth first traversal in which vertices are <u>discovered and fully explored</u>, starting from vertex (1) and use the numerical order to break ties, is:**
   a. 1, 2, 3, 6, 5, 4
   b. 4, 5, 6, 3, 2, 1
   c. 1, 2, 3, 4, 5, 6
   d. 2, 3, 6, 5, 4, 1

10. **Which of the following paths represents Hamiltonian Cycle?**
    a. 1, 2, 3, 1
    b. 1, 2, 3, 6, 5, 4, 1
    c. 1, 2, 3, 6, 5, 4, 1, 3
    d. 1, 3, 6, 5, 4, 1

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Question 2: Graph — To be answered in (15) Minutes [ ] / 5.5 Marks

A. Use undirected-weighted graph below to find a minimum spanning tree (MST) by using *Kruskal algorithm*. (1) draw the resulting minimum spanning tree, (2) specify the value of MCST and (3) specify the order in which edges get added to the spanning tree?



The resulting minimum spanning tree:

The value of MCST:

The *order* in which edges get added to the spanning tree:

B. If Prim's algorithm is applied, what is the *order* in which vertices get added to the spanning tree (i.e., removed from the priority queue)? Start with vertex *0*.

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Question 3: Sorting To be answered in (10) Minutes [ ] / 2.5 Marks

Using <u>min heap</u>, apply the heapsort algorithm on the array A= {30, 54, 12, 4, 40}. Show the sorting details by (1) draw the min heap that represents array A and (2) draw each sorting step by showing the tree and the corresponding array.

Min heap:

The heapsort steps:

Tree	Array

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Question 4: To be answered in (5) Minutes [ ] / 2 Marks

Consider the Kruskal's algorithm below.

```
MST-KRUSKAL(G, w)
1  A = ∅
2  for each vertex v ∈ G.V
3       MAKE-SET(v)
4  sort the edges of G.E into nondecreasing order by weight w
5  for each edge (u, v) ∈ G.E, taken in nondecreasing order by weight
6       if FIND-SET(u) ≠ FIND-SET(v)
7            A = A ∪ {(u, v)}
8            UNION(u, v)
9  return A
```

a) Assume that edge weights are not significantly greater than the number of edges, modify the Kruskal's algorithm in more efficient way?

b) What is the overall time complexity of the modified version of the algorithm. Show the details.